

1. How cleaner.perl works:

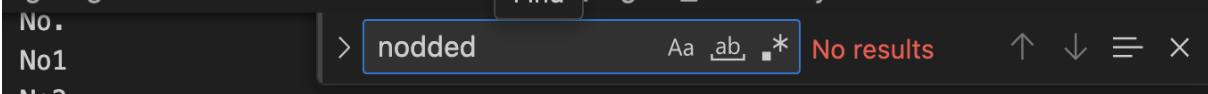
- a. Read \$fh line by line.
- b. For each line, convert it to an array called @tokens.
- c. Apply Regex to remove any non-alphanumeric characters before the word using '^'.
- d. Apply Regex to remove any non-alphanumeric characters after the word using '\$'.
 - i. We can select non-alphanumeric characters using the ^ (not) operator on a-zA-Z0-9 (All upper/lowercase letters and digits).
- e. Convert each word to lowercase using the lc() built-in function.

2. Running Script with Dictionary.txt:

```
misspelled word 213: wasn't
misspelled word 214: lefebvre-yoshida
misspelled word 215: wasn't
misspelled word 216: hmpph
misspelled word 217: it's
misspelled word 218: don't
misspelled word 219: don't
```

We can see that our script paired with spell_check.perl incorrectly detects contractions (not in our dictionary) but does pick up words that do not exist in the english language like lefebvre-yoshida.

3. Running Script with Oxford Dictionary

A screenshot of a search interface. On the left, there is a list of items labeled 'No.', 'No1', and 'No2'. To the right of this list is a search bar containing the text 'nodded'. To the right of the search bar is a filter dropdown menu showing 'Aa' and 'ab', followed by a search icon and the text 'No results'. To the right of the search bar are four navigation icons: an up arrow, a down arrow, a hamburger menu, and a close button.

```
misspelled word 3878: nodded
misspelled word 3879: abstractedly
misspelled word 3880: cuties
misspelled word 3881: signal
```

We can now see that we are detecting words that are spelled correctly but they are not in our dictionary which indicates our script is working as intended. There are fewer words that are being detected. I will note I had to update the original spell_check.perl as it was not set up correctly (not converting to lowercase and trimming whitespace).

4. Answers to Questions

Explain in your own words how **spell_check.perl** works.

- a. `Spell_check.perl` first reads in a file (dictionary) line by line and converts each line to lowercase and stores the cleaned input as a key paired with a 1. This indicates that the word was cleaned.
- b. The script then processes the command line arguments to see what text we need to compare with our cleaned dictionary. It implements proper error handling.
- c. It then opens the file, cleans it by removing whitespaces, and iterates over each word in a `@words` array and checks if it exists in the given dictionary. If it does, no handling happens, but if not, we increment our `count_misspelled` variable and print the value to the screen.
- d. The files are closed.

Explain in your own words how did you implement `cleaner.perl`.

- a. I stored each token in a `@tokens` array in my `while ($fh)` loop.
- b. I iterated over each element in the `tokens` array and applied the following regex:
 - i. `s/` = substitute
 - ii. `^` = start of word
 - iii. `[^a-zA-Z0-9]+//g` selects all non-alphanumeric characters before the word.
 - iv. I repeated the process except with a `$` at the end to check for non-alphanumeric characters after a word.
 - v. I printed the `lc` (lowercase) value of the word to the provided file.

Explain in your own words how did you implement `oxford_spell_check.perl`.

- a. I renamed the file to `oxford_spell_check.perl`.
- b. I replaced all regex lowercase detection to `lc()` so that it is compatible with UTF-8.
- c. I replaced the dictionary name with the Oxford Dictionary that I fetched using `wget`.
- d. I created an executable so that I can run with `./` using `chmod +x`.
- e. Renamed the file to `[file].txt` instead of `csv` since no files were comma-separated.

How many unique words you identified as "misspelled words" after performing Step 3?

- a. 986.

How many unique words you identified as "misspelled words" after performing Step 4?

- a. 4888.

How can you fix all misspelled words? suggest some solutions.

- a. You can fix this in a few ways. You can collect a larger dictionary, use an Oxford API, or use a Natural Language Processing (NLP) model to detect.